COURSE 3 · PHYSICS

Physics

Printable workbook with worked solutions

72 problems across 9 modules, plus the course capstone.
Full worked solutions at the back.

Name _____ Started _____

Before you start

How to use this workbook

There are 72 problems, eight per module, followed by the course capstone. Dots show difficulty: one dot is a warm-up, two is the standard you are aiming for, three stretches towards first-year university.

Work in pencil and show your steps. Try each problem before looking at the solutions, which start on the page headed *Worked solutions*. The same problems are in the interactive workbook in the Second Pass app, which checks answers instantly. The app also has a quiz for every module and course, and spaced-repetition flashcards; a printable set of the flashcards comes with each course.

A good routine for each module

Problems 1–2 straight after listening to the reader. Problems 3–6 after the explorer. Problems 7–8 at the end of the week, then take the module quiz in the app and redo anything you got wrong without looking. Review the module's flashcards for a few minutes each day after that.

Module 3.1

Motion (kinematics)

3.1.1 ●○○

A cyclist accelerates from 2 m/s to 8 m/s in 4 s. What is her acceleration?

Answer: _____ m/s^2

3.1.2 ●○○

A stone is dropped from rest. How fast is it falling after 3 s? ($g = 9.8\ m/s^2$)

Answer: _____ m/s

3.1.3 ●●○

How far does that stone fall in 3 s?

Answer: _____ m

3.1.4 ●●○

A car travelling at 25 m/s brakes at 5 m/s^2. What is its stopping distance?

Answer: _____ m

3.1.5 ●●○

A ball is thrown straight up at 14 m/s. How high does it go?

Answer: _____ m

3.1.6 ●●○

A ball rolls off a 1.25 m high table at 3.0 m/s. How far from the table does it land?

Answer: _____ m

3.1.7 ●●●

A projectile is launched at 20 m/s at 30° above level ground. What is its range?

Answer: _____ m

3.1.8 ●●○

At the top of a ball's flight, its acceleration is

A. zero

B. 9.8 m/s^2 downwards

C. 9.8 m/s^2 upwards

D. changing direction

Module 3.2

Forces and Newton's laws

3.2.1 ●○○

What net force gives a 1200 kg car an acceleration of 2.5 m/s^2?

Answer: _____ N

3.2.2 ●○○

What is the weight of a 65 kg person on Earth ($g = 9.8\ m/s^2$)?

Answer: _____ N

3.2.3 ●●○

A 10 kg box is pushed with 50 N across a floor where friction is 20 N. What is its acceleration?

Answer: _____ m/s^2

3.2.4 ●●○

A 5 kg block sits on a frictionless 30° slope. What is its acceleration down the slope?

Answer: _____ m/s^2

3.2.5 ●●○

A crate of mass 40 kg rests on the floor. The coefficient of friction is 0.30. What horizontal force is needed to start it sliding?

Answer: _____ N

3.2.6 ●●○

A 0.5 kg ball on a 0.8 m string is swung in a horizontal circle at 4 m/s. What is the tension (ignore gravity)?

Answer: _____ N

3.2.7 ●●○

A book rests on a table. The reaction partner (Newton's third law) of the book's weight is

A. the table's push on the book

B. the book's gravitational pull on Earth

C. friction

D. the table's weight

3.2.8 ●●●

A 70 kg person stands in a lift accelerating upwards at 1.5 m/s^2. What does a bathroom scale under them read, in newtons?

Answer: _____ N

Module 3.3

Energy, work and power

3.3.1 ●○○

How much work is done lifting a 20 kg box 1.5 m? (g = 9.8)

Answer: _____ J

3.3.2 ●○○

What is the kinetic energy of a 1000 kg car at 20 m/s?

Answer: _____ J

3.3.3 ●●○

A ball is dropped from 5.0 m. Ignoring air resistance, how fast is it moving when it hits the ground?

Answer: _____ m/s

3.3.4 ●●○

A 60 kg climber gains 300 m of height in 20 minutes. What is her average power output against gravity?

Answer: _____ W

3.3.5 ●●○

A 2.0 kW kettle runs for 3 minutes. How many joules does it use?

Answer: _____ J

3.3.6 ●●○

A motor uses 500 J of electrical energy to lift a load, gaining 350 J of potential energy. What is its efficiency?

Answer: _____ %

3.3.7 ●●○

A rope pulls a sled 10 m with 80 N at 30° to the ground. How much work is done?

Answer: _____ J

3.3.8 ●●●

If a car's speed rises from 50 km/h to 100 km/h, by what factor does its braking distance increase (same braking force)?

Answer: _____ ×

Module 3.4

Momentum, collisions and rotation

3.4.1 ●○○

What is the momentum of a 0.16 kg cricket ball moving at 30 m/s?

Answer: _____ kg m/s

3.4.2 ●●○

A 1500 kg car at 20 m/s hits a stationary 1000 kg car and they lock together. What is their speed just after?

Answer: _____ m/s

3.4.3 ●●○

A 0.06 kg tennis ball arrives at 20 m/s and leaves in the opposite direction at 30 m/s. What impulse did the racket give?

Answer: _____ N s

3.4.4 ●●○

If that contact lasted 0.005 s, what was the average force?

Answer: _____ N

3.4.5 ●●○

Airbags reduce injury mainly because they

A. reduce momentum change

B. increase collision time

C. absorb momentum permanently

D. increase mass

3.4.6 ●●○

A 30 N force acts at right angles to a spanner 0.25 m from the bolt. What is the torque?

Answer: _____ N m

3.4.7 ●●○

A 4.0 kg rifle fires a 0.010 kg bullet at 600 m/s. What is the rifle's recoil speed?

Answer: _____ m/s

3.4.8 ●●●

A spinning skater pulls in her arms, halving her moment of inertia. Her rotation rate

A. halves

B. doubles

C. is unchanged

D. quadruples

Module 3.5

Waves and sound

3.5.1 ●○○

A wave has frequency 50 Hz and wavelength 4 m. What is its speed?

Answer: _____ m/s

3.5.2 ●○○

What is the period of a 250 Hz sound?

Answer: _____ s

3.5.3 ●●○

Sound travels at 340 m/s. What is the wavelength of a 440 Hz note?

Answer: _____ m

3.5.4 ●●○

You see lightning and hear thunder 3 s later. Sound travels at 340 m/s. How far away was the strike?

Answer: _____ m

3.5.5 ●●○

A guitar string 0.65 m long carries waves at 420 m/s. What is its fundamental (n = 1) frequency?

Answer: _____ Hz

3.5.6 ●●○

Sound passes from air into water, where it travels faster. Its frequency

A. increases

B. decreases

C. stays the same

D. becomes zero

3.5.7 ●●○

Sound waves are

A. transverse

B. longitudinal

C. electromagnetic

D. standing only

3.5.8 ●●●

A 700 Hz siren approaches at 20 m/s. Sound speed is 340 m/s. What frequency do you hear?

Answer: _____ Hz

Module 3.6

Electricity and circuits

3.6.1 ●○○

A current of 2 A flows for 30 s. How much charge passes?

Answer: _____ C

3.6.2 ●○○

A 12 V supply drives current through a 4 Ω resistor. What is the current?

Answer: _____ A

3.6.3 ●●○

Resistors of 3 Ω, 5 Ω and 12 Ω are connected in series. What is the total resistance?

Answer: _____ Ω

3.6.4 ●●○

A 6 Ω and a 12 Ω resistor are connected in parallel. What is the combined resistance?

Answer: _____ Ω

3.6.5 ●●○

A 230 V heater draws 8 A. What is its power?

Answer: _____ W

3.6.6 ●●○

A 60 W lamp runs on 230 V. What is its resistance when lit?

Answer: _____ Ω

3.6.7 ●●○

A 2.4 kW appliance runs for 2.5 hours. Electricity costs 30 cents per kWh. What does it cost, in dollars?

Answer: _____ dollars

3.6.8 ●●●

Power is transmitted at higher voltage mainly to

A. increase current

B. reduce current and so I^2R heating losses

C. make it safer at home

D. increase resistance

Module 3.7

Magnetism and electromagnetism

3.7.1 ●○○

A 0.5 m wire carrying 4 A lies at right angles to a 0.2 T field. What force acts on it?

Answer: _____ N

3.7.2 ●●○

A transformer has 500 primary turns and 25 secondary turns. With 240 V in, what is the output voltage?

Answer: _____ V

3.7.3 ●●○

An electron ($q = 1.6 \times 10^{-19}$ C) moves at 2×10^{6} m/s perpendicular to a 0.01 T field. What force acts on it?

Answer: _____ N

3.7.4 ●●○

A magnet is held still inside a coil. The induced current is

A. large

B. zero

C. alternating

D. growing

3.7.5 ●●○

Who first showed that an electric current produces a magnetic field?

A. Faraday

B. Ørsted

C. Maxwell

D. Newton

3.7.6 ●●○

A coil of 200 turns sees its flux change by 0.03 Wb in 0.1 s. What size of voltage is induced?

Answer: _____ V

3.7.7 ●●○

Transformers do not work with steady direct current because

A. DC is too strong

B. there is no changing magnetic field

C. DC has no voltage

D. coils resist DC

3.7.8 ●●●

Explain in a few sentences how a bicycle dynamo lights a lamp.

Module 3.8

Thermodynamics and heat

3.8.1 ●○○

Convert 25 °C to kelvin.

Answer: _____ K

3.8.2 ●○○

How much energy raises 2.0 kg of water by 10 K? ($c = 4200$ J/kg K)

Answer: _____ J

3.8.3 ●●○

How much energy melts 0.5 kg of ice at 0 °C? ($L = 3.34 \times 10^5$ J/kg)

Answer: _____ J

3.8.4 ●●○

A 2 kW kettle heats 1.5 kg of water from 20 °C to 100 °C. Ignoring losses, how long does it take?

Answer: _____ s

3.8.5 ●●○

A gas gains 500 J of heat and does 200 J of work on its surroundings. What is the change in its internal energy?

Answer: _____ J

3.8.6 ●●○

What is the maximum efficiency of an engine running between 600 K and 300 K?

Answer: _____ %

3.8.7 ●●○

A refrigerator moves heat from cold to hot. Does this break the second law?

A. Yes

B. No, because work is supplied and more entropy is produced outside

C. No, because fridges are cold

D. Only in summer

3.8.8 ●●●

Explain why steam at 100 °C causes worse burns than water at 100 °C.

Module 3.9

Modern physics

3.9.1 ●○○

How much energy is equivalent to 1 g of mass? ($c = 3 \times 10^8$ m/s)

Answer: _____ J

3.9.2 ●●○

What is the energy of a photon of green light, frequency 5.6×10^{14} Hz? ($h = 6.63 \times 10^{-34}$ J s)

Answer: _____ J

3.9.3 ●●○

A radioactive isotope has a half-life of 8 days. What fraction remains after 24 days? Give a decimal.

Answer: _____

3.9.4 ●●○

A sample of wood has 25% of its original carbon-14. How old is it? (half-life 5730 years)

Answer: _____ years

3.9.5 ●●●

A spaceship travels at 0.8c. By what factor do its clocks run slow (γ)?

Answer: _____

3.9.6 ●●○

In the photoelectric effect, increasing the brightness of light below the threshold frequency

A. ejects more electrons

B. ejects faster electrons

C. ejects no electrons

D. lowers the threshold

3.9.7 ●●○

The Sun produces energy mainly by

A. fission of uranium

B. fusion of hydrogen into helium

C. chemical burning

D. radioactive decay

3.9.8 ●●●

Why does each element produce its own pattern of spectral lines?

Capstone

A satellite's journey

A 500 kg weather satellite is carried to orbit and released. Use motion, forces, energy and modern physics to work out key numbers for its mission. Take g at the surface as 9.8 m/s^2, Earth's radius as 6.37×10^6 m, GM for Earth as 3.99×10^{14} m^3/s^2.

Part 1 ●●○

What is the satellite's weight on the launch pad?

Answer: _____ N

Part 2 ●●○

In a circular orbit at radius r, gravity supplies the centripetal force: $GMm/r^2 = mv^2/r$. Find the orbital speed at $r = 7.0 \times 10^6$ m.

Answer: _____ m/s

Part 3 ●●○

What is its kinetic energy in that orbit?

Answer: _____ J

Part 4 ●●○

Its solar panels deliver 1.2 kW. How much energy do they supply in one 100-minute orbit?

Answer: _____ J

Part 5 ●●●

Its radio transmits at 2.0×10^9 Hz. What is the wavelength? ($c = 3.0 \times 10^8$ m/s)

Answer: _____ m

Part 6 ●●●

The satellite's clock is compared with clocks on the ground. Name the two relativistic effects that make them drift apart.

Answers

Worked solutions

Module 3.1 · Motion (kinematics)

3.1.1 ANSWER 1.5 m/s²

1 a = (v − u)/t.

2 (8 − 2) ÷ 4 = 1.5 m/s².

3.1.2 ANSWER 29.4 m/s

1 v = u + at = 0 + 9.8 × 3.

3.1.3 ANSWER 44.1 m

1 s = ut + ½at².

2 ½ × 9.8 × 9 = 44.1 m.

3.1.4 ANSWER 62.5 m

1 v² = u² + 2as with v = 0, a = −5.

2 0 = 625 − 10s, s = 62.5 m.

3.1.5 ANSWER 10 m

1 At the top v = 0: 0 = 14² − 2 × 9.8 × s.

2 s = 196 ÷ 19.6 = 10 m.

3.1.6 ANSWER 1.515 m

1 Vertical: 1.25 = ½ × 9.8 × t², so t ≈ 0.505 s.

2 Horizontal: 3.0 × 0.505 ≈ 1.5 m.

3.1.7 ANSWER 35.35 m

1 R = u² sin 2θ / g.

2 400 × sin 60° ÷ 9.8 ≈ 35.3 m.

3.1.8 ANSWER B. 9.8 m/s² downwards

1 Gravity still acts. Velocity is momentarily zero, not acceleration.

Module 3.2 · Forces and Newton's laws

3.2.1 ANSWER 3000 N

1 F = ma = 1200 × 2.5.

3.2.2 ANSWER 637 N

1 $W = mg = 65 \times 9.8$.

3.2.3 ANSWER 3 m/s²

1 Net force = 50 − 20 = 30 N.

2 $a = 30 \div 10 = 3$ m/s².

3.2.4 ANSWER 4.9 m/s²

1 Component along slope: $mg \sin\theta$.

2 $a = g \sin 30° = 4.9$ m/s².

3.2.5 ANSWER 117.6 N

1 $N = mg = 392$ N.

2 $F = \mu N = 0.30 \times 392 \approx 118$ N.

3.2.6 ANSWER 10 N

1 $F = mv^2/r$.

2 $0.5 \times 16 \div 0.8 = 10$ N.

3.2.7 ANSWER B. the book's gravitational pull on Earth

1 The weight is Earth pulling the book. Its partner is the book pulling Earth.

3.2.8 ANSWER 791 N

1 $N - mg = ma$.

2 $N = 70(9.8 + 1.5) = 791$ N.

Module 3.3 · Energy, work and power

3.3.1 ANSWER 294 J

1 $W = mgh$.

2 $20 \times 9.8 \times 1.5 = 294$ J.

3.3.2 ANSWER 2 × 10⁵ J

1 $\frac{1}{2} \times 1000 \times 400 = 200\,000$ J.

3.3.3 ANSWER 9.899 m/s

1 $mgh = \frac{1}{2}mv^2$, so $v = \sqrt{(2gh)}$.

2 $\sqrt{(98)} \approx 9.9$ m/s.

3.3.4 ANSWER 147 W

1 Energy = mgh = 176 400 J.

2 ÷ 1200 s ≈ 147 W.

3.3.5 ANSWER 3.6×10^5 J

1 E = Pt = 2000 × 180.

2 = 360 000 J.

3.3.6 ANSWER 70 %

1 350 ÷ 500 = 0.70.

3.3.7 ANSWER 692.8 J

1 W = Fd cos θ.

2 80 × 10 × 0.866 ≈ 693 J.

3.3.8 ANSWER 4 ×

1 Braking work Fd must remove ½mv^2.

2 $d \propto v^2$, so doubling v multiplies d by 4.

Module 3.4 · Momentum, collisions and rotation

3.4.1 ANSWER 4.8 kg m/s

1 p = mv = 0.16 × 30.

3.4.2 ANSWER 12 m/s

1 Momentum before = 30 000 kg m/s.

2 v = 30 000 ÷ 2500 = 12 m/s.

3.4.3 ANSWER 3 N s

1 Δp = m(v − u) = 0.06 × (30 − (−20)).

2 = 3.0 N s.

3.4.4 ANSWER 600 N

1 F = Δp/Δt = 3.0 ÷ 0.005 = 600 N.

3.4.5 ANSWER B. increase collision time

1 Same Δp over a longer time means smaller force.

3.4.6 ANSWER 7.5 N m

1 τ = Fr = 30 × 0.25.

3.4.7 ANSWER 1.5 m/s

1 Total momentum zero before.

2 4v = 0.01 × 600, v = 1.5 m/s.

3.4.8 ANSWER B. doubles

1 L = Iω is conserved.

2 Halve I, double ω.

Module 3.5 · Waves and sound

3.5.1 ANSWER 200 m/s

1 v = fλ = 50 × 4.

3.5.2 ANSWER 0.004 s

1 T = 1/f = 1/250.

3.5.3 ANSWER 0.7727 m

1 λ = v/f = 340 ÷ 440 ≈ 0.77 m.

3.5.4 ANSWER 1020 m

1 Light arrives almost instantly.

2 d = 340 × 3 = 1020 m.

3.5.5 ANSWER 323.1 Hz

1 $f_1 = v/2L$.

2 420 ÷ 1.3 ≈ 323 Hz.

3.5.6 ANSWER C. stays the same

1 Frequency is set by the source; wavelength changes instead.

3.5.7 ANSWER B. longitudinal

1 Air molecules oscillate along the direction of travel.

3.5.8 ANSWER 743.8 Hz

1 $f' = f\,v/(v - v_s)$.

2 700 × 340 ÷ 320 ≈ 744 Hz.

Module 3.6 · Electricity and circuits

3.6.1 ANSWER 60 C

1 Q = It = 2 × 30.

3.6.2 ANSWER 3 A

1 I = V/R = 12 ÷ 4.

3.6.3 ANSWER 20 Ω

1 Add them: 3 + 5 + 12.

3.6.4 ANSWER 4 Ω

1 1/R = 1/6 + 1/12 = 3/12.

2 R = 4 Ω.

3.6.5 ANSWER 1840 W

1 P = VI = 230 × 8.

3.6.6 ANSWER 881.7 Ω

1 $P = V^2/R$, so $R = V^2/P$.

2 52 900 ÷ 60 ≈ 882 Ω.

3.6.7 ANSWER 1.8 dollars

1 Energy = 2.4 × 2.5 = 6 kWh.

2 6 × $0.30 = $1.80.

3.6.8 ANSWER B. reduce current and so I^2R heating losses

1 Same P = VI with higher V means lower I.

2 Loss I^2R falls with the square of current.

Module 3.7 · Magnetism and electromagnetism

3.7.1 ANSWER 0.4 N

1 F = BIL = 0.2 × 4 × 0.5.

3.7.2 ANSWER 12 V

1 Vs = Vp × Ns/Np = 240 × 25/500.

3.7.3 ANSWER 3.2×10^{-15} N

1 F = qvB.

2 $1.6 \times 10^{-19} \times 2 \times 10^{6} \times 0.01 = 3.2 \times 10^{-15}$ N.

3.7.4 ANSWER B. zero

1 Induction needs a changing flux.

3.7.5 ANSWER B. Ørsted

1 Ørsted, 1820.

3.7.6 ANSWER 60 V

1 $\varepsilon = N\Delta\Phi/\Delta t = 200 \times 0.03 \div 0.1 = 60$ V.

3.7.7 ANSWER B. there is no changing magnetic field

1 Induction requires change.

3.7.8 ANSWER The wheel spins a magnet near a coil (or a coil near a magnet). The magnetic flux through the coil keeps changing, which by Faraday's law induces a voltage. That voltage drives a current through the lamp. The faster you ride, the faster the flux changes, so the lamp glows brighter.

1 What is changing, and what does that cause?

Module 3.8 · Thermodynamics and heat

3.8.1 ANSWER 298 K

1 25 + 273.

3.8.2 ANSWER 8.4e+04 J

1 $Q = mc\Delta T = 2 \times 4200 \times 10$.

3.8.3 ANSWER 1.67×10^5 J

1 $Q = mL = 0.5 \times 3.34 \times 10^5 = 1.67 \times 10^5$ J.

3.8.4 ANSWER 252 s

1 $Q = 1.5 \times 4200 \times 80 = 504\,000$ J.

2 $t = Q/P = 252$ s.

3.8.5 ANSWER 300 J

1 $\Delta U = Q - W = 500 - 200$.

3.8.6 ANSWER 50 %

1 $1 - 300/600 = 0.5$.

3.8.7 ANSWER B. No, because work is supplied and more entropy is produced outside

1 The second law forbids this happening spontaneously, not with work input.

3.8.8 ANSWER When steam condenses on skin it releases its latent heat of vaporisation, about 2.26 million joules per kilogram, before it even starts cooling as water. Water at 100 °C only releases heat as it cools. So the same mass of steam delivers far more energy to the skin.

1 Think about latent heat.

Module 3.9 · Modern physics

3.9.1 ANSWER 9×10^{13} J

1 $E = mc^2 = 0.001 \times 9 \times 10^{16}$.

2 $= 9 \times 10^{13}$ J.

3.9.2 ANSWER 3.71×10^{-19} J

1 $E = hf = 6.63 \times 10^{-34} \times 5.6 \times 10^{14} \approx 3.7 \times 10^{-19}$ J.

3.9.3 ANSWER 0.125

1 24 ÷ 8 = 3 half-lives.

2 $(½)^3 = 0.125$.

3.9.4 ANSWER 1.146e+04 years

1 25% = $(½)^2$, two half-lives.

2 2 × 5730 = 11 460 years.

3.9.5 ANSWER 1.667

1 $\gamma = 1/\sqrt{(1 - 0.64)} = 1/\sqrt{0.36}$.

2 $= 1/0.6 \approx 1.67$.

3.9.6 ANSWER C. ejects no electrons

1 Each photon must individually have enough energy; more low-energy photons don't help.

3.9.7 ANSWER B. fusion of hydrogen into helium

1 Hydrogen fusion in the core.

3.9.8 ANSWER Electrons in an atom can only occupy certain energy levels, and the spacing of those levels depends on the element's nuclear charge and electron arrangement. When an electron drops between levels it emits a photon with energy exactly equal to the gap, $E = hf$, so only certain frequencies appear. Different elements have different level spacings and so different line patterns.

1 Link energy levels to photon frequency.

Capstone · A satellite's journey

Part 1 ANSWER 4900 N

1 $W = mg = 500 \times 9.8$.

Part 2 ANSWER 7550 m/s

1 $v = \sqrt{(GM/r)}$.

2 $\sqrt{(3.99 \times 10^{14} \div 7.0 \times 10^{6})} \approx 7550$ m/s.

Part 3 ANSWER 1.43×10^{10} J

1 $\frac{1}{2}mv^2 = \frac{1}{2} \times 500 \times (7550)^2$.

2 $\approx 1.4 \times 10^{10}$ J.

Part 4 ANSWER 7.2×10^{6} J

1 $E = Pt = 1200 \times 6000$ s.

2 $= 7.2 \times 10^{6}$ J.

Part 5 ANSWER 0.15 m

1 $\lambda = c/f = 3 \times 10^{8} \div 2 \times 10^{9} = 0.15$ m.

Part 6 ANSWER Special relativity: the satellite moves fast relative to the ground, so its clock runs slow (time dilation). General relativity: it sits higher in Earth's gravitational field, where clocks run faster. For GPS satellites the gravitational effect wins, and both must be corrected for the system to work.

1 One effect comes from speed, the other from gravity.